Rajnil Malviya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.71
Sanskar Valley Public HS School	Class XII (Board of Secondary Education , Madhya Pradesh)	2022	85.00%
Sanmati HS School	Class X (Central Board of Secondary Education)	2020	87.60%

WORK EXPERIENCE

Machine Learning Intern

LightSpeak.AI

- Developed an end-to-end AI assistant for real-estate queries, enabling users to interact with property information through natural-language conversations.

- Built a **RAG-based question-answering system** using **ChromaDB**, enabling context-aware responses and efficient retrieval of relevant information.

- Integrated **Sarvam AI** speech-to-text and text-to-speech, enabling seamless voice-based interactions with the assistant.

TECHNICAL SKILLS

Languages : C++ , Python , C , MATLAB

ML / AI : RAG, Agentic AI, PyTorch, TensorFlow, LLMs, Computer Vision, Prompt Engineering

Backend / Databases : FastAPI, PostgreSQL, Redis, pgvector, SQL

Tools / Infrastructure : Docker, AWS, Git, Linux, OpenCV

PROJECTS

Atlas - Autonomous AI Agent and ML Copilot

Python, FastAPI, LLMs, PostgreSQL, Redis, pgvector, Docker, AWS

Built an end-to-end **autonomous AI agent platform from scratch**, capable of understanding complex requests, planning multi-step workflows, executing tools, and synthesizing results into natural-language responses.

Developed a robust **agent execution framework** supporting task dependencies, parallel execution, execution state, retries, timeouts, feedback-driven recovery, and extensible tool integration for complex workflows.

Implemented **persistent episodic and semantic memory** with PostgreSQL, pgvector, and Redis, enabling Atlas to retain user context, retrieve relevant knowledge, and deliver personalized interactions across sessions.

Built a secure **multi-user AI platform** with authentication, persistent user/session data, **sandboxed Python shell**

and code execution, isolated runtime environments, concurrency controls, and AWS deployment.
Integrated **Google applications, web access, web scraping, read/write operations, and Python execution** through an extensible tool ecosystem, enabling Atlas to perform real-world actions and workflows.
Designed **token-aware context management and structured LLM prompting** for reliable planning and response generation; currently extending Atlas into an **ML Copilot** for end-to-end machine-learning workflows.

Biped Vision Robot - Video Summarization

PyTorch, OpenCV, Docker

Built a lightweight **CNN-MLP based video summarization system** for a biped robot, enabling long visual streams to be condensed into concise segments containing salient events.
Developed a learned **visual importance estimation pipeline** for automatically identifying important video content using **TVSum**.
Designed the system to preserve **continuous events and activities** in generated summaries, validated it on unseen SUMMe videos achieving a **0.27 F1-score**, and containerized the inference pipeline using **Docker**.

DSP Project - Real-Time Noise Cancellation using RLS

MATLAB

Implemented real-time **noise cancellation system** using the **Recursive Least Squares (RLS)** algorithm.
Enabled **selective ambient noise removal** with a dual-mode interface (full/partial filtering).
Achieved **Signal-to-Noise Ratio (SNR) gain of 40 dB** in synthetic and real-world test scenarios.
Tested algorithm on multiple audio samples to evaluate clarity and convergence behavior.

RELEVANT COURSES

Computer Science :- Computer Architecture , Data Structures , Algorithms , Object-Oriented Programming
Machine Learning :- Multi Agent Sysrems , Linear Regression Models , Linear Classification Models , KNN
Mathematics :- Optimization Techniques , Statistical Inference , Random Processes, Calculus
Electrical Engineering :- Information Theory , Communication Systems , Digital Signal Processing

POSITION OF RESPONSIBILITY

Multimedia Head - Extra Mural Lectures Club , IIT Hyderabad
Multimedia Coordinator - Sunshine , IIT Hyderabad

EXTRACURRICULAR

Served as Deputy Coordinator for IIT Hyderabad at the Inter IIT Cultural Meet 2025 hosted by IIT Kanpur.
Represented IIT Hyderabad in Street Play and Stand-up Comedy at Inter IIT Cultural Meet 2024, held at IIT Patna.
Participated in the Filmmaking Competition as part of the Inter-Hostel Cultural Fest at IIT Hyderabad.